

Lecture 6

–

Visual EDA

(Exploratory Data Analysis)

Star / Starling

(*Sturnus vulgaris*)

- It is about 20 cm long and has glossy black plumage with a metallic sheen, which is speckled with white at some times of year. The legs are pink and the bill is black in winter and yellow in summer.
- It is a noisy bird, especially in communal roosts and other gregarious situations, with an unmusical but varied song. Its gift for mimicry has been noted in literature including the the works of William Shakespeare.
- The common starling is a highly gregarious species, especially in autumn and winter. Although flock size is highly variable, huge, noisy flocks may form near roosts.
- Due to numerous naturalizations on other continents, the starling is now one of the most common birds in the world.
- The star was "Bird of the Year 2018" in Germany.

Sources:
* Photo by Сергей Шабанов on pixabay
* https://en.wikipedia.org/wiki/Common_starling

Focusing Objectives

By the end of today's lecture, you will be able to:

1. Use Pandas, Matplotlib, and Seaborn for visual EDA
2. Create univariate, bivariate, and multivariate plots
3. Visually identify outliers, missing values, and trends



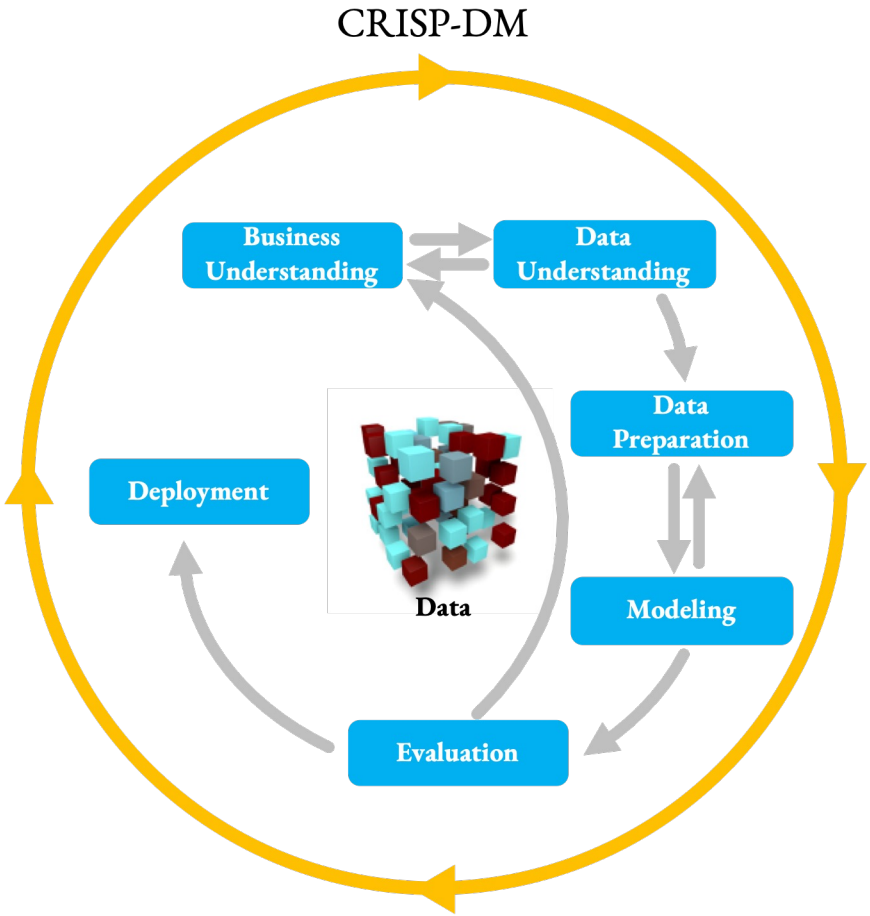
Visual EDA

1. CRISP-DM: Data Understanding – Visual EDA
2. Univariate Visualizations
3. Bivariate Visualizations
4. Multivariate Visualizations

Data Understanding and Preparation Phases of CRISP-DM

Data Understanding

- Main goal: gain a comprehensive understanding of the data that will be used for the data mining project.
- Data quality is being analyzed.



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What is Exploratory Data Analysis (EDA)?

Exploratory Data Analysis refers to process of

summarizing, visualizing, and interpreting datasets to extract insights before modelling

so as
- to discover patterns,
- to understand distributions of variables,
- to identify relationships between variables,
- to spot anomalies (outliers, missing values),
- to test hypothesis and
- to make informed decisions about data cleaning and feature selection

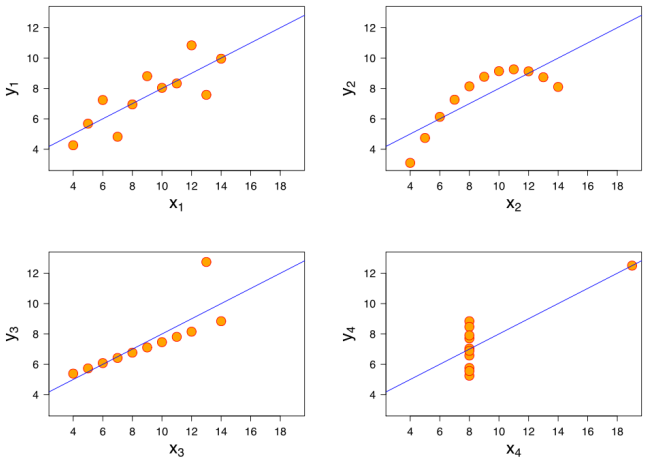
with the help of summary statistics and graphical representations.

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Why is Visual EDA Important?

- Raw data can be misleading: Visualizing helps identify patterns
  - Easier to spot trends and outliers than using raw numbers
    - Example: Anscombe's Quartet
  - Improves model accuracy by understanding data distributions

- Summary Statistics & Visual EDA work together
  - Summary Stats: Get quick numerical insights ("`describe()`"), but misses patterns, skewness, anomalies
  - Visual EDA: Shows distributions, but hard to interpret for high-dimensional data



Anscombe's quartet: four datasets that have nearly identical simple descriptive statistics

source: wikipedia.org

(Pandas and) Matplotlib and Seaborn

- “Pandas” for quick visualizations
- “Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.”

  See https://matplotlib.org/
- “Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.” “It builds on top of matplotlib and integrates closely with pandas data structures.”

  See https://seaborn.pydata.org/

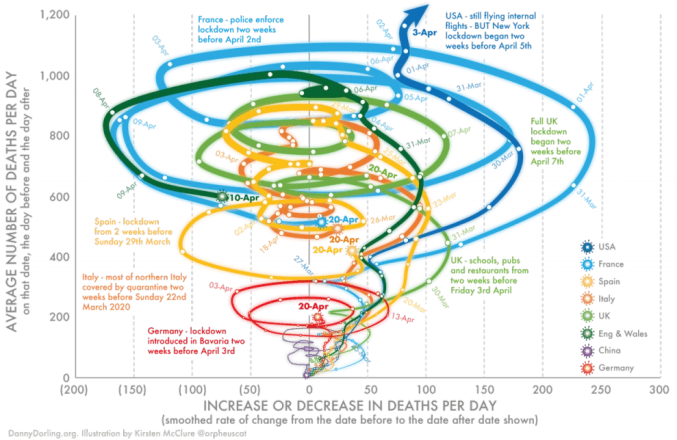


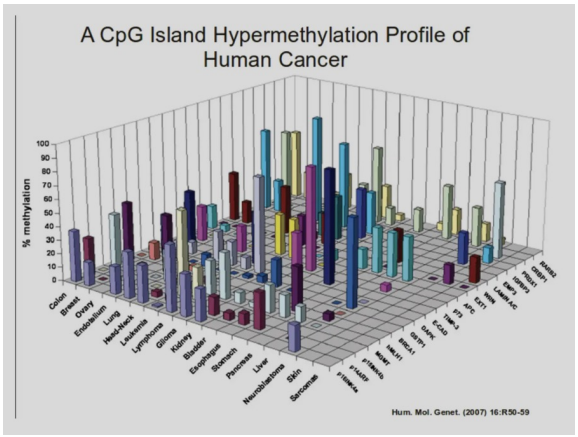


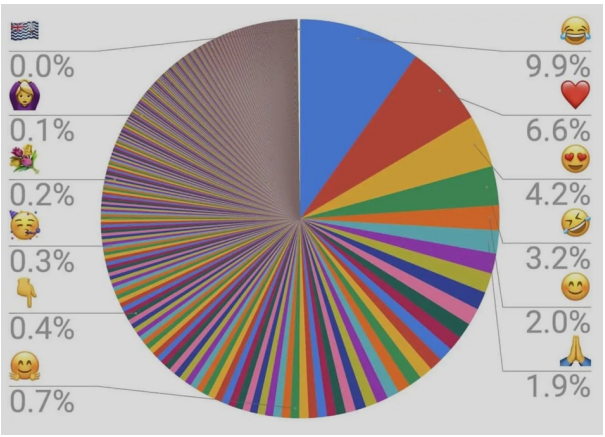
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Caveats

- Many visualizations have been proposed
  - And easy to generate – powerful libraries available
- Many are visually pretty, but very hard to interpret
  - Use visualizations that are easy to interpret
  - You will need them for communication with domain experts from the business side later
  - We only look at the most widely used visualizations for EDA
- Exploring data is fun!
  - Do not get carried away
  - Focus on what to explore
  - Spend your time wisely!







Source: https://www.oldstreetsolutions.com/good-and-bad-data-visualization

Visual EDA

1. CRISP-DM: Data Understanding – Visual EDA
2. Univariate Visualizations
3. Bivariate Visualizations
4. Multivariate Visualizations

What is Univariate Analysis?

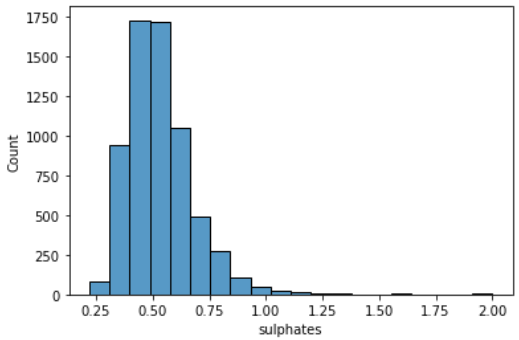
- Definition: examine one variable at a time
- Key Questions Answered
  - What is the distribution of the variable?
  - Are there any outliers?
  - Is the data skewed or normally distributed?
- Data Types of Variables
  - Numerical (continuous & discrete)
    - Use Histograms & KDE Plots
    - Use Boxplots, Violinplots
  - Categorical (nominal & ordinal)
    - Use Frequency/Count Plots
- Choosing the right visualization depends on the data type and research question
- Do not use: Pie Charts, any 3D Chart → too hard to correctly interpret for Humans

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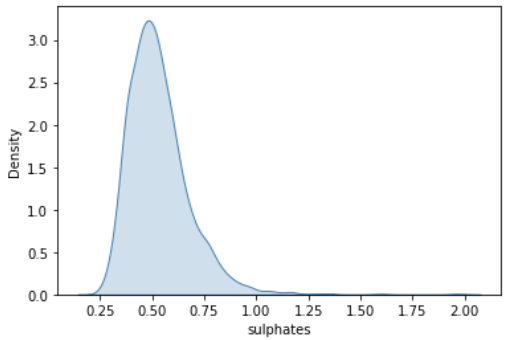
kernel density estimation

Numerical Univariate Data – Histograms & KDE Plots

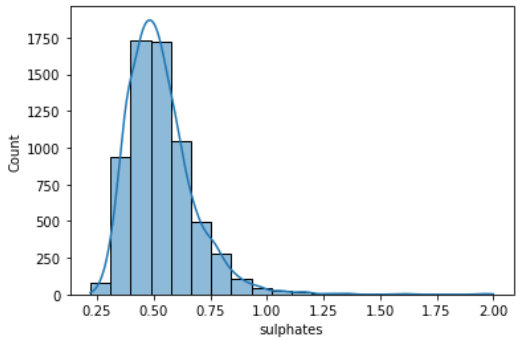
- Histogram: Groups data into bins to show frequency distribution
- Density Plots: smooth version of a histogram that estimates data density
- Use cases
  - Detect skewness, modality (uni- / bi- / multimodal distributions)
  - Understand if data follows a normal, uniform, or skewed distribution



Histogram



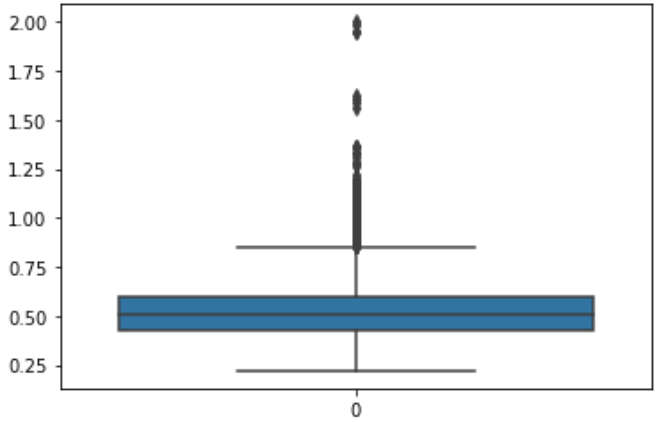
Density Plot



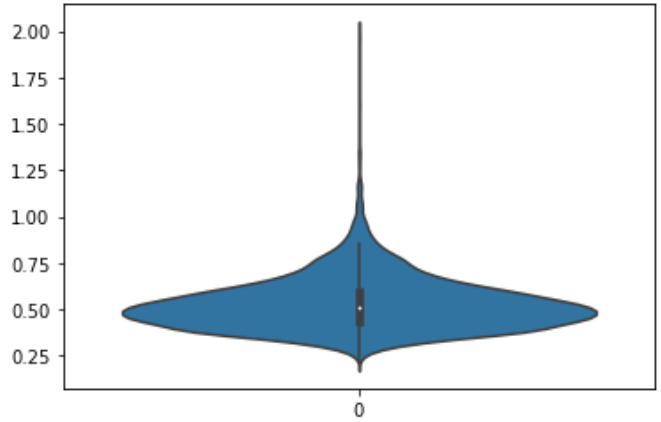
Combined Histogram and Density Plot

Numerical Univariate Data – Boxplots and Violinplots

- Box Plots: Shows median, quartiles (Q1, Q3), whiskers, and outliers
- Violin Plots: Like Boxplots, but showing the KDE instead of the box
- Use cases
  - Identify outliers
  - Compare distributions across categories



Box Plot

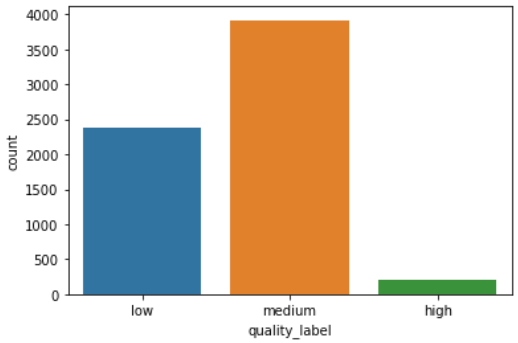


Violin Plot

© Markus Breunig

Categorical Univariate Data – Frequency/Count Plots

- Frequency/Count Plot: Shows frequency/count of categorical values
- Use cases
  - Compare category sizes in a dataset
  - Class label: evaluate if balanced or imbalanced classes



Frequency Plot

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Demo

Demo

Visual EDA (Univariate)



Source: Foto von Markus Spiske auf Unsplash

Exercise

Exercise 1

Iris Data – Univariate Visualizations for EDA

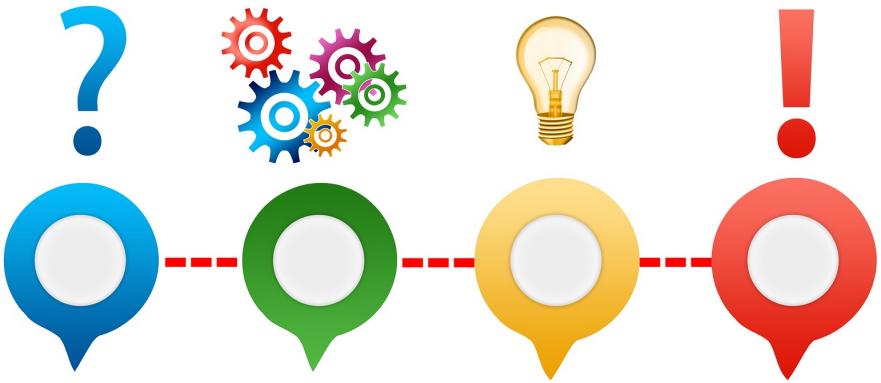


Photo by Gerd Altmann on Pixabay

Visual EDA

1. CRISP-DM: Data Understanding – Visual EDA
2. Univariate Visualizations
3. Bivariate Visualizations
4. Multivariate Visualizations

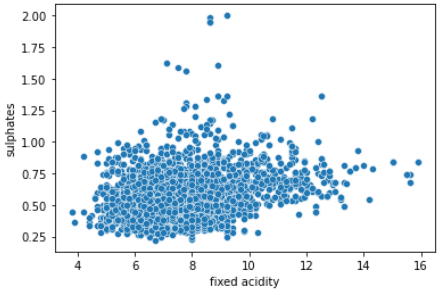
What is Bivariate Analysis?

- Definition: examine the relationship between two variables
- Key Questions Answered
  - Are two variables correlated?
  - Does one variable influence another?
  - Are there patterns or trends in the relationship?
- Types of Bivariate Relationships:
  - Numerical-Numerical
    - Use Scatter Plots / Joint Plots
    - Use Correlation Heatmaps
  - Categorical-Categorical
    - Use Multi Bar Plots
  - Categorical-Numerical
    - Use Multi Boxplots & Multi Violinplots
    - Use Strip Plots & Swarm Plots

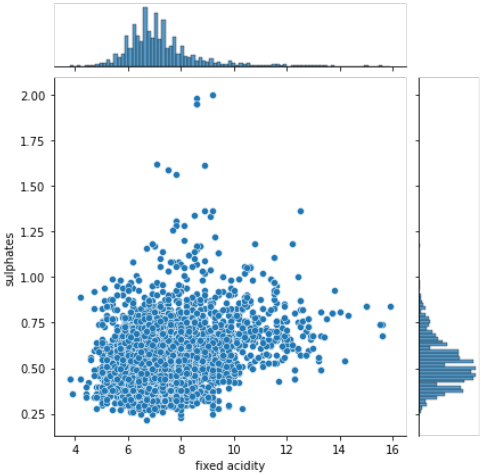
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Numerical Bivariate Data – Scatter Plots & Joint Plots

- Scatter Plot
  - Shows the relationship between two numerical variables
  - Each point represents an observation
- Joint Plot
  - Add Histograms/KDE Plots on the sides
- Use cases
  - Identify positive, negative, or no correlation
  - Detect clusters and outliers



Scatter Plot

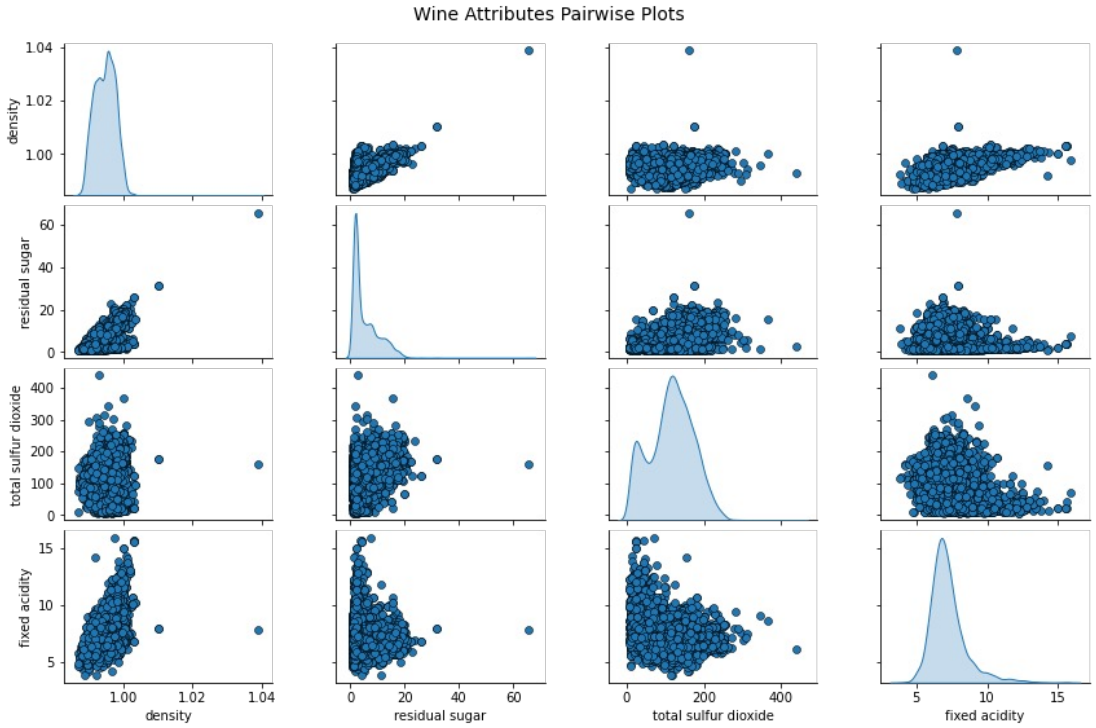


Joint Plot

© Markus Breunig

Numerical Bivariate Data – Pair-Wise Scatter Plots

- Pair-Wise Scatter Plots
  - Show all combinations & KDE Plots
- Use Cases
  - Get a quick overview for small dataset with not-too-many numerical attributes
  - Very expensive for large datasets and many numerical attributes!!



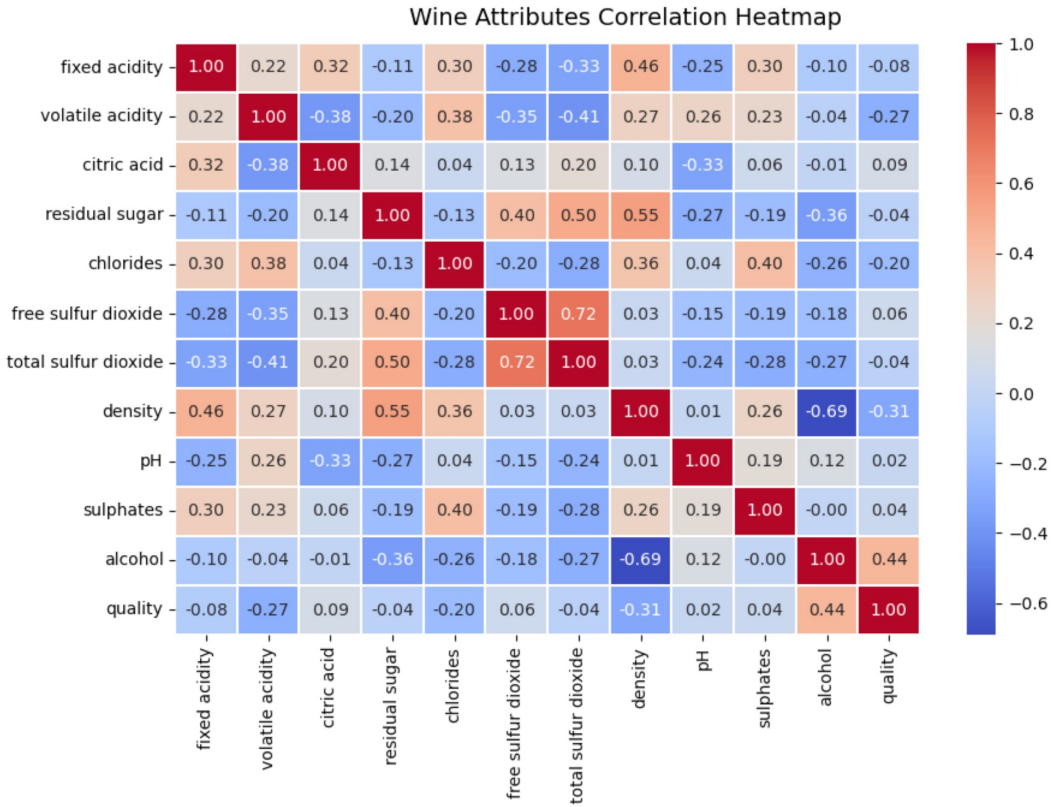
Numerical Bivariate Data – Correlation Heatmap

- Correlation Heatmap
  - Measures how strongly two numerical variables are related.
  - Most frequently based on Pearson or Spearman Correlation
  - Often coded from Blue to Red (`coolwarm`-Colour Map)

- Use cases
  - Quickly spot relationships between multiple variables
  - Helps in feature selection for machine learning

pearson: linear correlation
> 1 means both increase linearly (positive correlation),
i.e. if one attribute goes up, the other goes up as well (same amount = 1)
> -1 mean both decrease linearly (negative correlation)

both very red and very blue are interesting



Numerical Bivariate Data – Correlation Coefficients for Two Variables

- Pearson Correlation Coefficient (r)
  - Measures how **linearly related** two variables are
  - Use it when:
    - You suspect a linear relationship
    - Data is continuous, roughly normally distributed, and free from extreme outliers
    - pandas: `df.corr(method='pearson')` (default)
- Spearman Rank Correlation Coefficient (ρ or rs)
  - Measures how well the relationship can be described by a **monotonic function**
    - Consistently increasing or decreasing, not necessarily linear
  - Use it when:
    - You expect a monotonic (but not necessarily linear) relationship
    - Data has outliers, is ordinal, or is not normally distributed
    - pandas: `df.corr(method='spearman')`

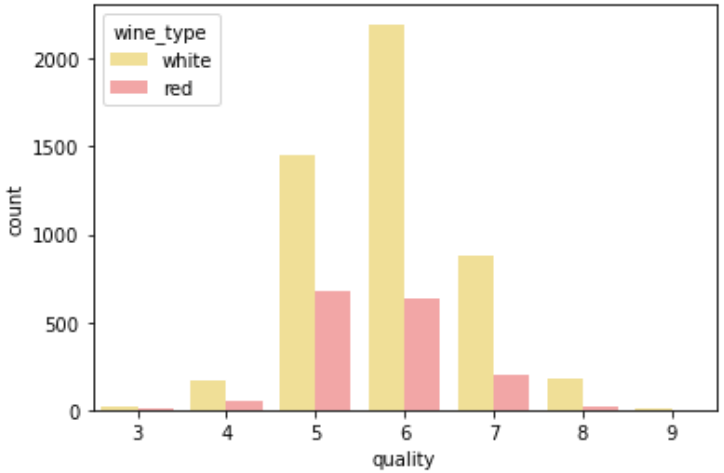
Feature	Pearson	Spearman
Type of relationship measured	Linear relationships	Monotonic relationships (increasing or decreasing)
Based on	Raw data values	Ranks of the data
Sensitivity to outliers	High	Low
Assumes	Normally distributed, continuous variables	No assumption of distribution
Range of values	[-1;1]	[-1;1]
Interpretation	1 = perfect positive linear 0 = no linear relationship -1 = perfect negative linear	Same, but for monotonic relationships

start with pearson
if smth strange, then also use spearman
generally, pearson works good on gaussian distribution, spearman otherwise

Categorical Bivariate Data – Multi Bar Plot

- Multi Bar Plot (Multi Count Plot, Count Plot, Multi Frequency Plot)
  - Shows the frequency of categories split by another category
- Use cases
  - Compare one distribution across another distribution

too many plots make it hard to read and interpret



Multi Bar Plot

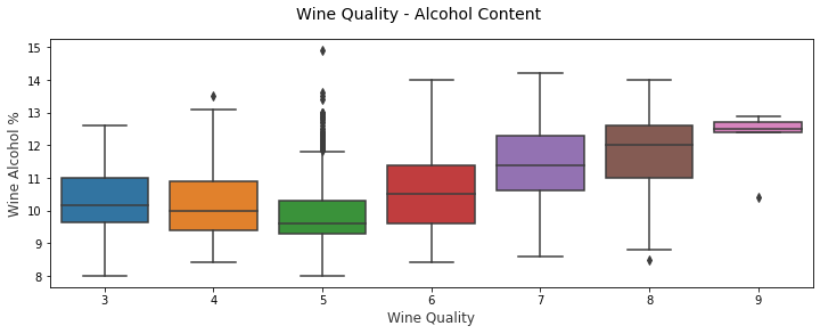
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Numerical-Categorical Bivariate Data – Multi Box Plot & Multi Violin Plot

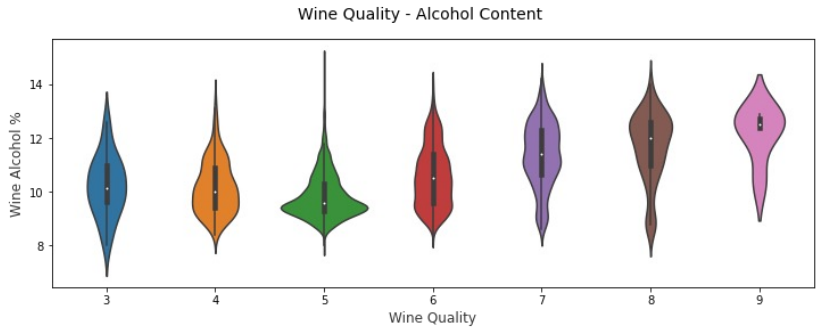
- Multi Box Plot: Shows the distribution of a numerical variable across categories
- Use cases
  - Compare distributions of numerical data across categories
  - Identify outliers

- Multi Violin Plot: Combines boxplot with KDE plot for richer insights
- Use cases
  - Identify data distribution within each category
  - Detect skewness and multimodal distributions

higher alcohol content > better wine quality



Multi Box Plot



Multi Violin Plot

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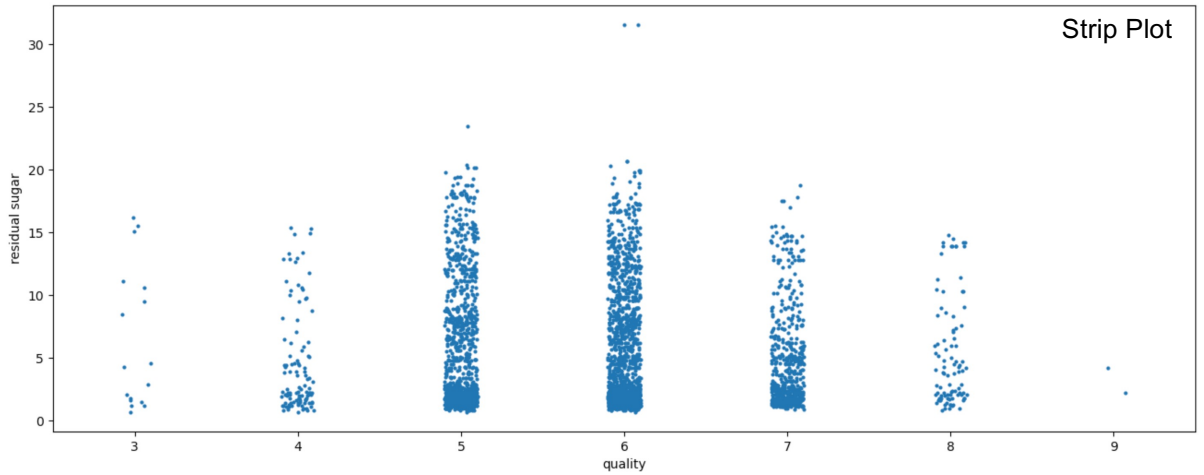
x axis is class label

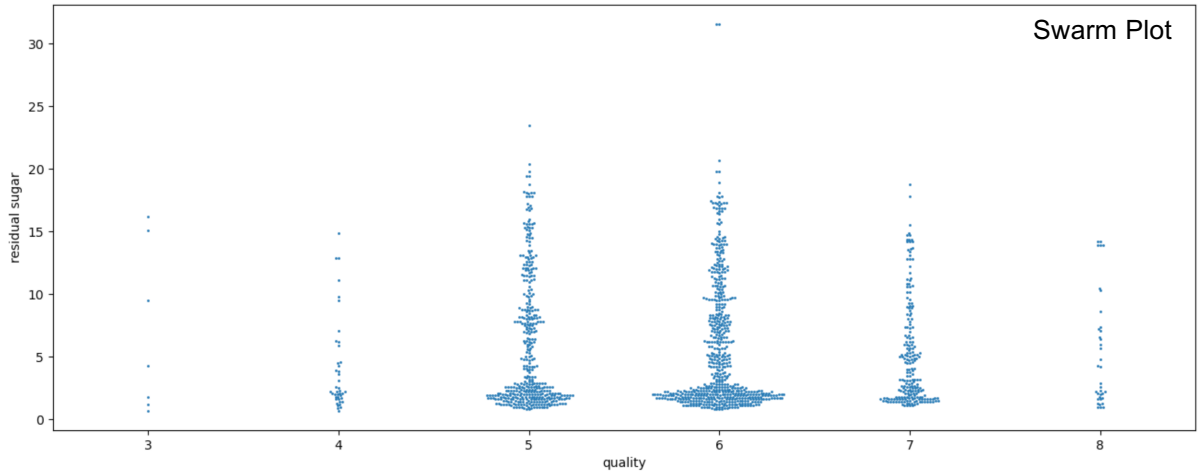
Numerical-Categorical Bivariate Data – Strip Plots & Swarm Plots

- Strip Plot
  - Categorical scatterplot using jitter to reduce overplotting
- Swarm Plot
  - Same but with non-overlapping points i.e. moves data points outward

  but necessitates enough space for all data points
- Use Cases
  - Spot overlapping values and outliers
- Note: the two graphs show two different 20% random samples

  as a result, the second graph has not data points for the wine quality nine





first and foremost, during data exploration all graphs are for oneself, so use the ones that are easiest to understand for oneself while carrying meaning

Visual EDA

1. Introduction
2. Univariate Visualizations
3. Bivariate Visualizations
4. Multivariate Visualizations

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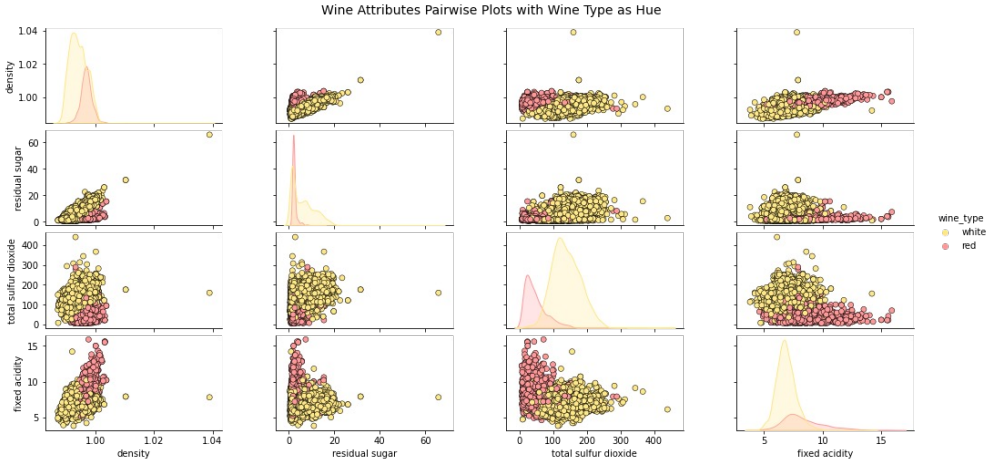
What is Multivariate Analysis?

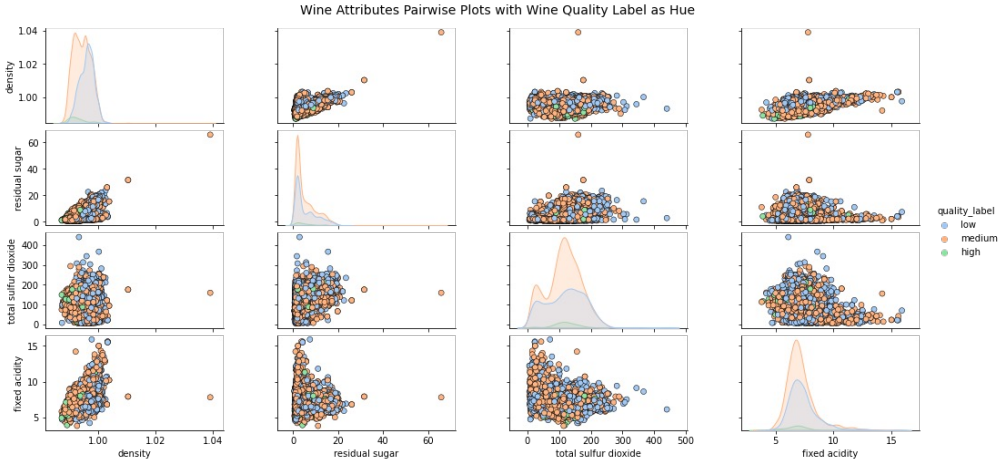
- Definition: examine the relationship between three or more variables
- Key Questions Answered
  - Are there complex patterns that bivariate analysis misses?
  - How do multiple variables interact?
  - Are there hidden relationships in the data?
- Often hard to implement and interpret!
- Approaches
  - Interactive visualizations can be helpful
    - Rotate a 3-d scatterplot
    - Time dimension as animation

    higher mental load since interacting and understanding correlations in 3D space is difficult
  - Introduce third dimension as colour / hue / size into bivariate visualizations
    - Frequently used for the target variable in classification problems

Multivariate – Pairwise Scatterplots with Hue as Third Dimension

- Pairwise Scatterplots with Hue as Third Dimension
  - Example 1: Hue for Wine Type
    - Ok
  - Example 2: Hue for Wine Quality
    - Hard to interpret

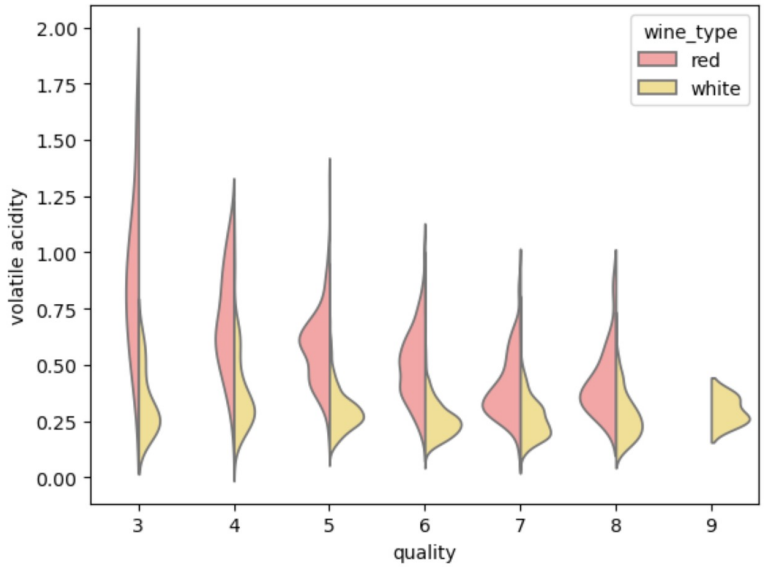




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Multivariate – Multi Violinplot with Hue for Binary Attribute

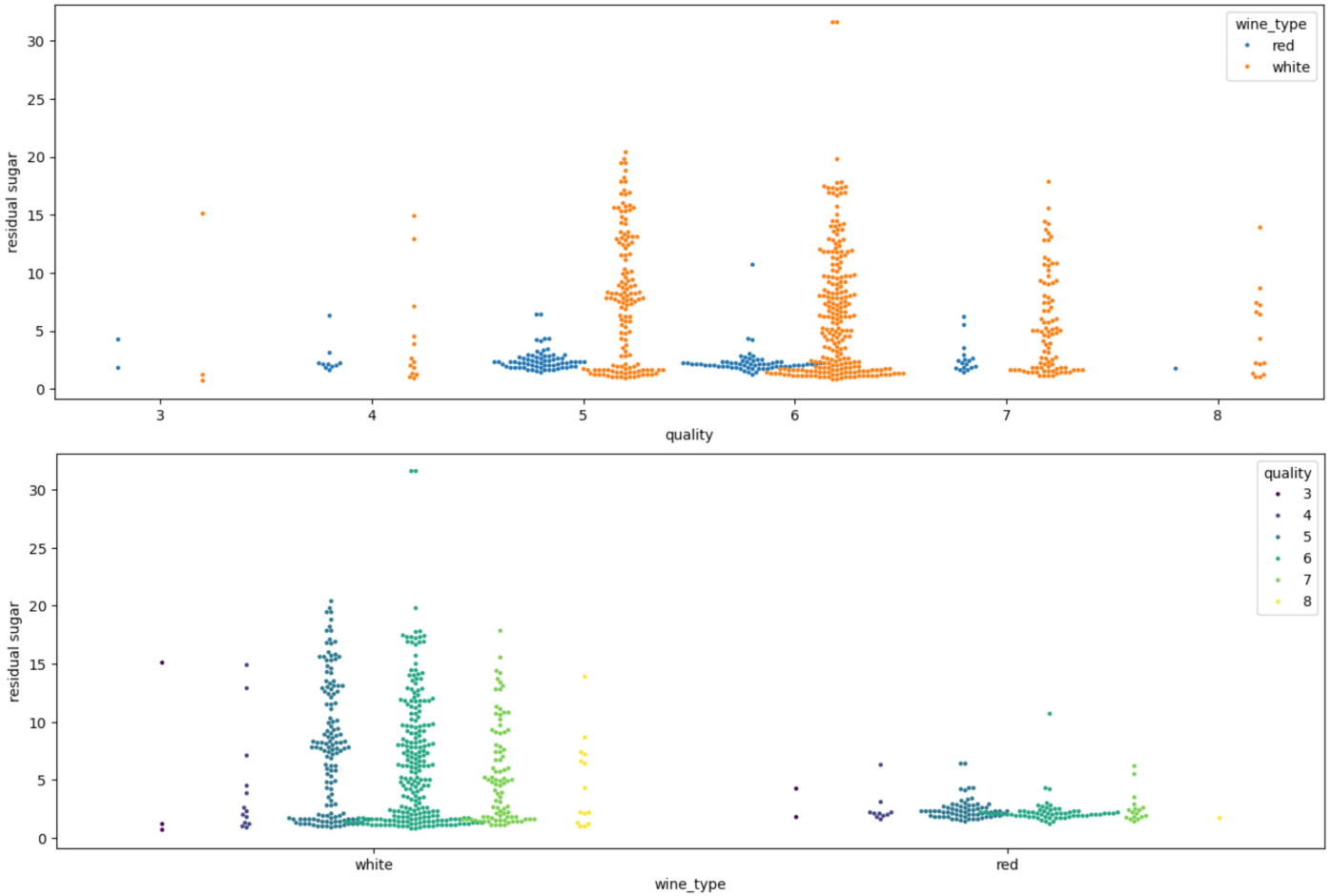
- Multi Violinplot with Hue for Binary Attribute
- Example
  - X: Quality (Categorical)
  - Y: Acidity (Numerical)
  - Hue: Wine Type (Binary)



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Multivariate – Swarm Plot with Hue as Third Dimension

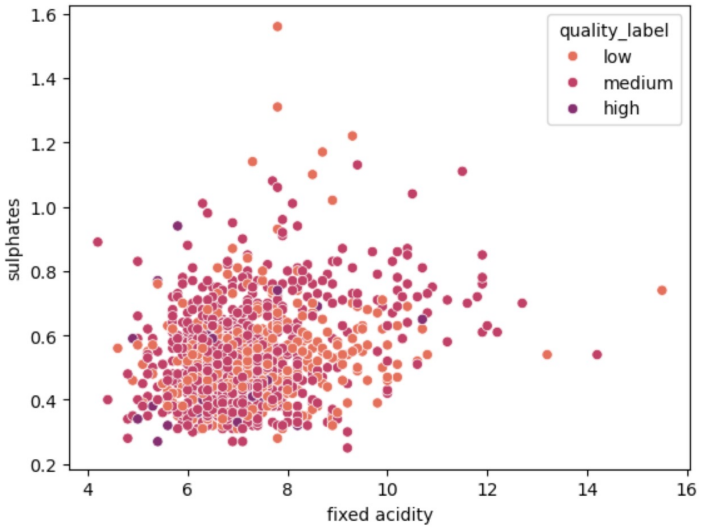
- Swarm Plot with Hue as Third Dimension
  - Example 1
    - X: Quality (Categorical)
    - Y: Sugar (Numerical)
    - Hue: Type (Binary)
  - Example 2:
    - X: Type (Binary)
    - Y: Sugar (Numerical)
    - Hue: Quality (Cat.)
- Note: the two graphs show two different 10% random samples



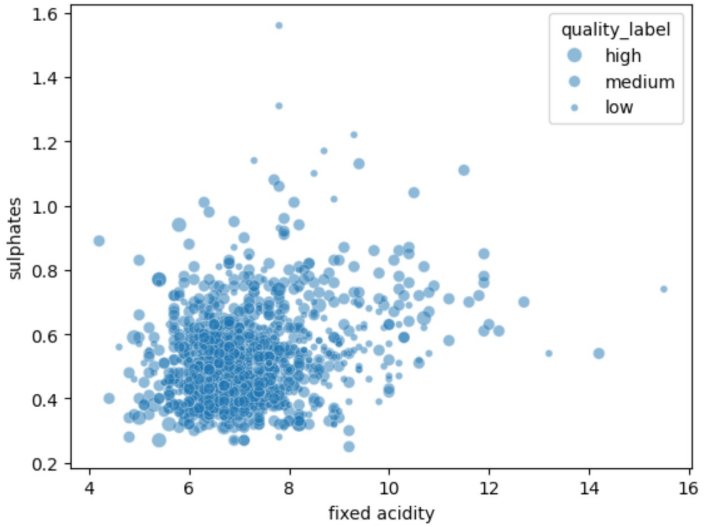
bottom graph not readable, too much at once

Multivariate – Scatterplot with Hue or Size as Third Dimension

- Scatterplot with Hue as Third Dimension
  - X: Numerical (fixed acidity)
  - Y: Numerical (sulpathes)
  - Hue: Categorical (quality_label)



- Scatterplot with Size as Third Dimension (Bubble Chart)
  - X: Numerical (fixed acidity)
  - Y: Numerical (sulpathes)
  - Size: Numerical or Categorical (here: categorical, quality_label)



Note: the two graphs show the same 20% random sample

negative conclusion here: see no clear distinction
note: negative conclusions can also be useful, exploration does not always have to yield only positive results

Demo

Demo

Visual EDA (Bi- and Multivariate)



Source: Foto von Markus Spiske auf Unsplash

Common Mistakes and Best Practices

- Common Mistakes in Visual EDA
  - Ignoring categorical variables when plotting numerical data
  - Using the wrong type of visualization (e.g., pie charts)
  - Not labelling axes or providing enough context for plots
- Some Best Practices for Effective Data Visualization
  - Choose the right visualization for your data type
  - Use appropriate labels, legends, and titles
  - Avoid misleading scales or truncation
  - Colour-code data effectively (e.g., divergent colours for negative vs. positive values)
  - Check for overplotting and adjust transparency (alpha parameter)

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Exercise

Exercise 2

Iris Data – Bivariate and Multivariate Visualizations for EDA

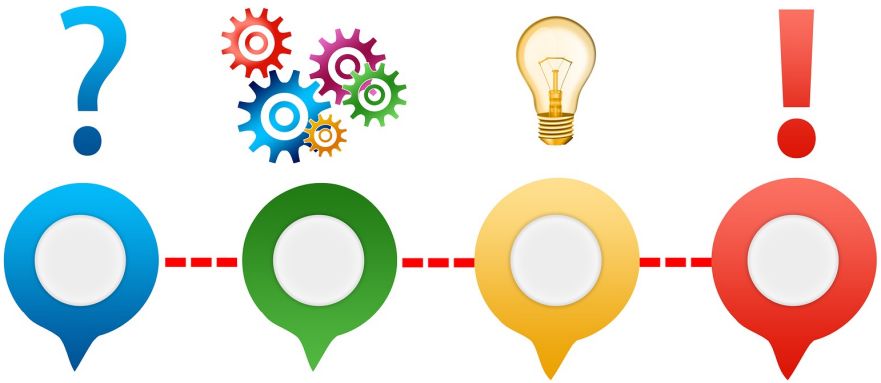


Photo by Gerd Altmann on Pixabay

Key Takeaways

- Visual Analysis of the data is key part of EDA
- Start with univariate visualizations of all features (and the target, if applicable)
  - Start with a Histogram/Density Plot/Frequency Plot
  - Look for Outliers in the Box Plots
- Continue with bivariate/multivariate visualizations
  - Pair-wise scatter plots
  - Colour code/hue code (the target), if applicable
- Use easy-to-interpret and easy-to-communicate visualizations
- Focus – explore deep but do not get carried away



Image by Gordon Johnson on pixabay